## Homework

### DM2 - Approximating a root - 50 min.

Consider $u_0 = 0$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \sqrt{2 + u_n}$. Questions 1, 2 and 4 are to be done after S20; question 3 will be revisited after the lesson on continuity in S31.

1. Prove by induction that all terms are defined and that $0 \leq u_n \leq 2$ for every $n \in \mathbb{N}$.

2. Prove by induction that $u_{n+1} \geq u_n$ for all $n \in \mathbb{N}$. Deduce that $(u_n)$ converges.

3. Using the continuity of the function $x \mapsto \sqrt{2 + x}$, determine the limit of $(u_n)$.

4. Prove that $0 \leq 2 - u_{n+1} \leq \frac{2 - u_n}{2}$ for all $n \in \mathbb{N}$. Deduce an upper bound for $2 - u_n$ in function of $n$, then an integer $N$ such that $0 \leq 2 - u_n \leq 10^{-3}$ for all $n \geq N$.